Recap: Bounded functions

Drawing a graph of a function without taking your pen off the paper led to the notion of a Darboux function.

Darboux $\not\Rightarrow$ Continuous.

Actual definition of continuous was $(\forall \varepsilon > 0)(\exists \delta > 0)(\forall x \in D)$
$(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)$

Theorem 4.1: Algebraic properties of continuous functions

(ii) Scratch Work:

Trying to prove that $fg: D \to \mathbb{R}$ is cts at $c \in D$. Recall, $(fg)(x) = f(x)g(x)$. Want:

$|f(x)g(x) - f(c)g(c)| < \varepsilon$, whenever $|x - c| < \delta$.

Know, $|f(x) - f(c)| < \varepsilon$ and $|g(x) - g(c)| < \varepsilon$.

Then, $|f(x)g(x) - f(c)g(c)| = |f(x)g(x) - f(x)g(c) + f(x)g(c) - f(c)g(c)|$

$$\leq |f(x)|\,|g(x)-g(c)| + |f(x)-f(c)|\,|g(c)|$$

$$< 1+|f(c)| \quad < \frac{\varepsilon}{2(1+|f(c)|)} \quad < \frac{\varepsilon}{2(|g(c)|+1)}$$

So, $|f(x)-f(c)| < 1 \cdot \big||f(x)|-|f(c)|\big| \leq |f(x)-f(c)|$

$$\Rightarrow \big||f(x)|-|f(c)|\big| < 1.$$

$$\Rightarrow |f(x)| < 1+|f(c)|.$$

Proof: Since f is cts at $c \in D$, it follows there exists $\delta_1 > 0$ s.t

$$|f(x)| < 1+|f(c)|$$

whenever $|x-c| < \delta_1$ and $x \in D$. Additionally, for every $\varepsilon > 0$, there exists $\delta_2 > 0$ s.t

$$|f(x)-f(c)| < \frac{\varepsilon}{2(|g(c)|+1)}$$

whenever $|x-c| < \delta_2$ and $x \in D$. Since g is cts at c, it follows there exists $\delta_3 > 0$ s.t

$$|g(x)-g(c)| < \frac{\varepsilon}{2(|f(c)|+1)}$$

whenever $|x-c| < \delta_3$ and $x \in D$. Define $\delta = \min(\delta_1, \delta_2, \delta_3)$.

Then if $|x-c| < \delta$ and $x \in D$, it follows that

$$|f(x)g(x) - f(c)g(c)| < |f(x)|\,|g(x)-g(c)| + |g(c)|\,|f(x)-f(c)|$$

$$< (|f(c)|+1)\,\frac{\varepsilon}{2(|f(c)|+1)} + |g(c)|\,\frac{\varepsilon}{2(|g(c)|+1)}$$

$$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$$

$$= \varepsilon. \qquad \square$$

(iii) See notes

(iv) Exercise.

Ex: Consider the function $p: \mathbb{R} \to \mathbb{R}$ given by

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$$

where $a_i \in \mathbb{R}$ for each $i \in \{0, 1, \dots, n\}$ and $n \in \mathbb{N} \cup \{0\}$.

These functions are called <u>polynomials of order n</u>. Since $f: \mathbb{R} \to \mathbb{R}$ given by $f(x) = x$ is cts $\forall c \in \mathbb{R}$ it follows

by the algebra of cts functions that p is cts $\forall c \in \mathbb{R}$.

Theorem: Let $f: D_1 \to \mathbb{R}$ and $g: D_2 \to \mathbb{R}$ with range $f(D_1) := \{f(x) : x \in D_1\}$ a subset D_2. Then if f is cts a $c \in D_1$, and g is cts at $f(c) \in f(D_1)$, then the composition $g \circ f$ is cts at $c \in D_1$.

Proof: Given $\varepsilon > 0$, there exists $\delta_1 > 0$ s.t

$$|g(x) - g(f(c))| < \varepsilon$$

whenever $x \in D_2$ and $|x - f(c)| < \delta_1$. Since $f(D_1) \subset D_2$ it follows that

$$|g(f(x)) - g(f(c))| < \varepsilon$$

whenever $f(x) \in D_2$ and $|f(x) - f(c)| < \delta_1$. Since f is cts at c, it follows there exists $\delta > 0$ s.t

$$|f(x) - f(c)| < \delta_1$$

whenever $|x - c| < \delta$ and $x \in D_1$. Hence, for every $\varepsilon > 0$

then exists $\delta > 0$ s.t

$$|g(f(x)) - g(f(c))| < \varepsilon$$

whenever $|x - c| < \delta$ and $x \in D_1$. That is, $g \circ f$ is cts a $c \in D_1$. ∎